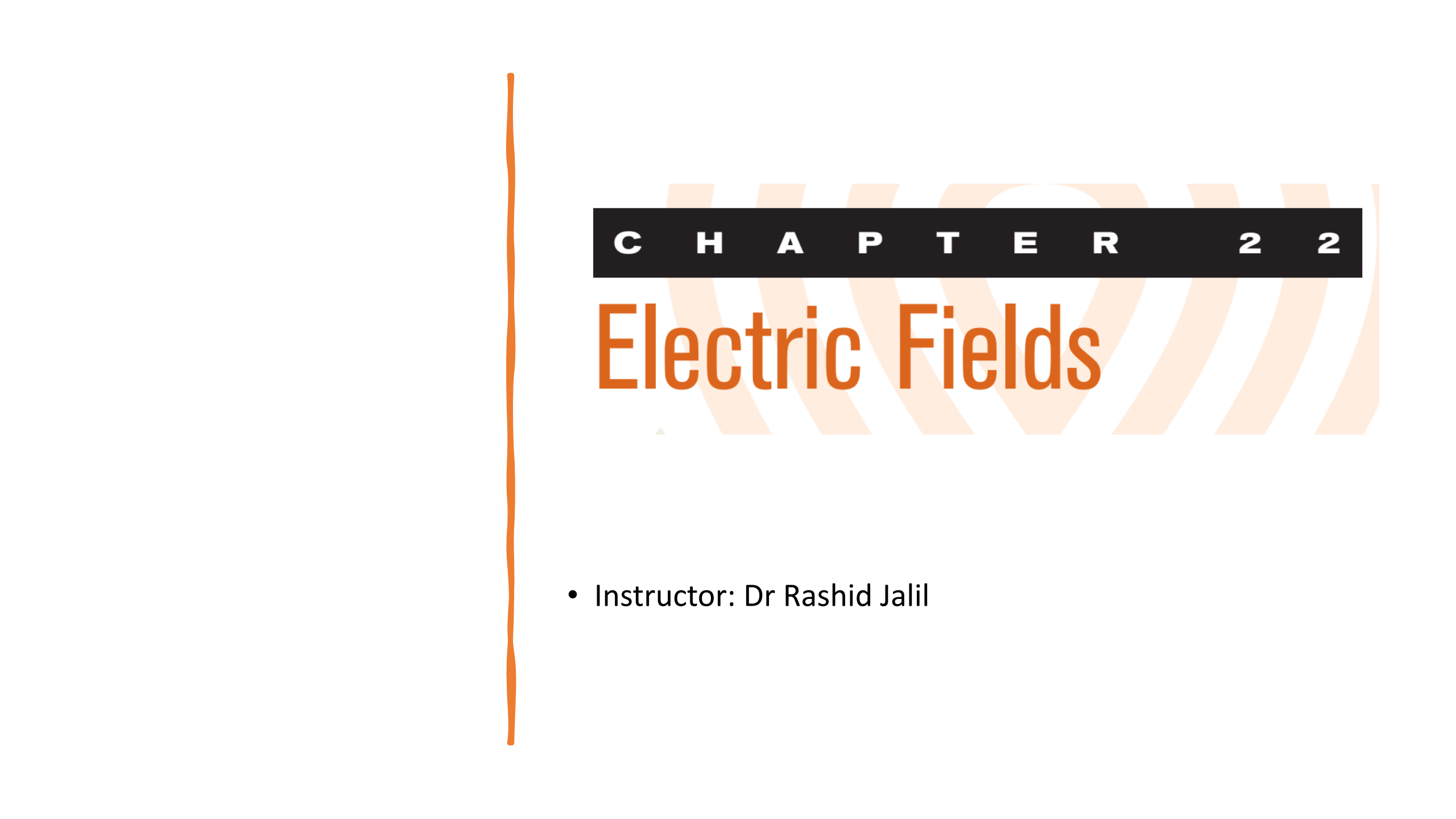




C H A P T E R 2 2

Electric Fields

- Instructor: Dr Rashid Jalil

22-1 THE ELECTRIC FIELD

Learning Objectives

After reading this module, you should be able to . . .

22.01 Identify that at every point in the space surrounding a charged particle, the particle sets up an electric field $\vec{E}$, which is a vector quantity and thus has both magnitude and direction.

22.02 Identify how an electric field $\vec{E}$ can be used to explain how a charged particle can exert an electrostatic force $\vec{F}$ on a second charged particle even though there is no contact between the particles.

22.03 Explain how a small positive test charge is used (in principle) to measure the electric field at any given point.

22.04 Explain electric field lines, including where they originate and terminate and what their spacing represents.

Key Ideas

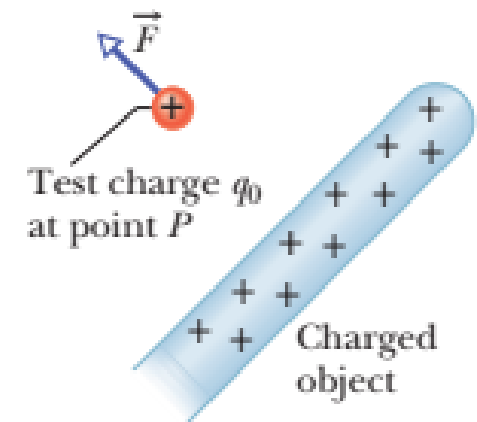
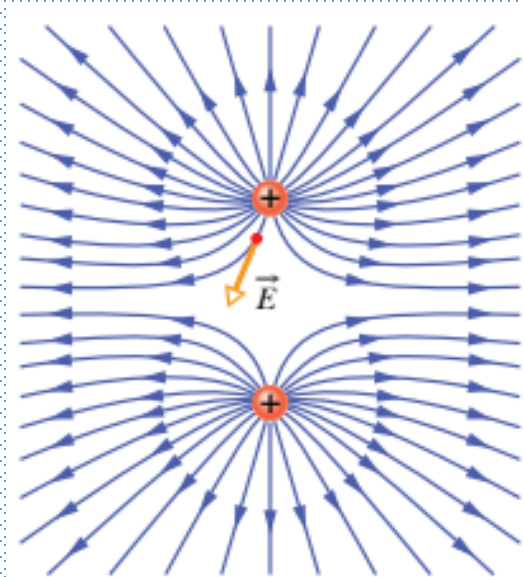
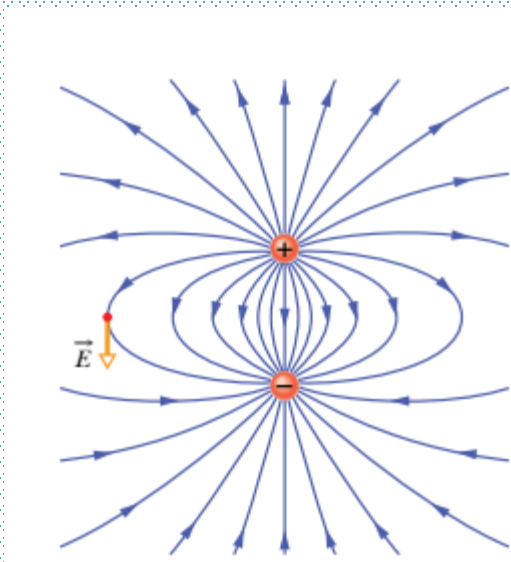
- A charged particle sets up an electric field (a vector quantity) in the surrounding space. If a second charged particle is located in that space, an electrostatic force acts on it due to the magnitude and direction of the field at its location.

- The electric field $\vec{E}$ at any point is defined in terms of the electrostatic force $\vec{F}$ that would be exerted on a positive test charge q_0 placed there:

$$\vec{E} = \frac{\vec{F}}{q_0}.$$

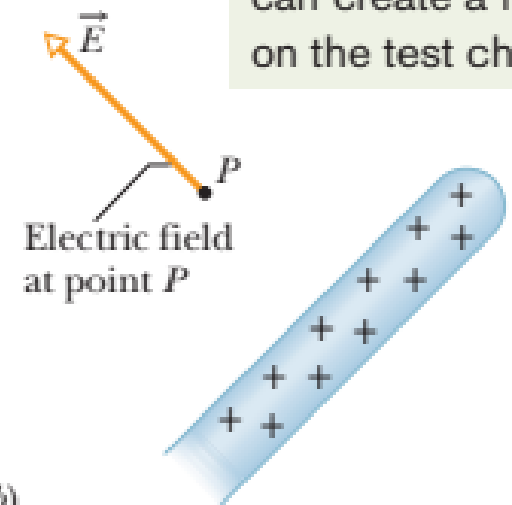
22-2 The Electric Field

$$\vec{E} = \frac{\vec{F}}{q_0} \quad (\text{electric field}). \quad (22-1)$$



(a)

The rod sets up an electric field, which can create a force on the test charge.



(b)

22-4 The Electric Field Due to a Point Charge

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2} \hat{i}. \quad (22-2)$$

$$\vec{E} = \frac{\vec{F}}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{i} \quad (\text{point charge}). \quad (22-3)$$

$$\vec{F}_0 = \vec{F}_{01} + \vec{F}_{02} + \cdots + \vec{F}_{0n}.$$

Therefore, from Eq. 22-1, the net electric field at the position of the test charge is

$$\begin{aligned} \vec{E} &= \frac{\vec{F}_0}{q_0} = \frac{\vec{F}_{01}}{q_0} + \frac{\vec{F}_{02}}{q_0} + \cdots + \frac{\vec{F}_{0n}}{q_0} \\ &= \vec{E}_1 + \vec{E}_2 + \cdots + \vec{E}_n. \end{aligned} \quad (22-4)$$

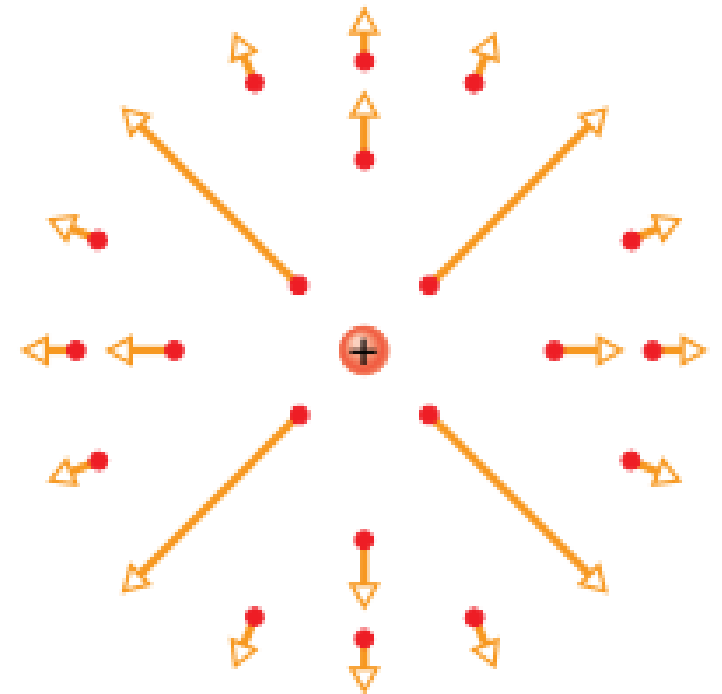
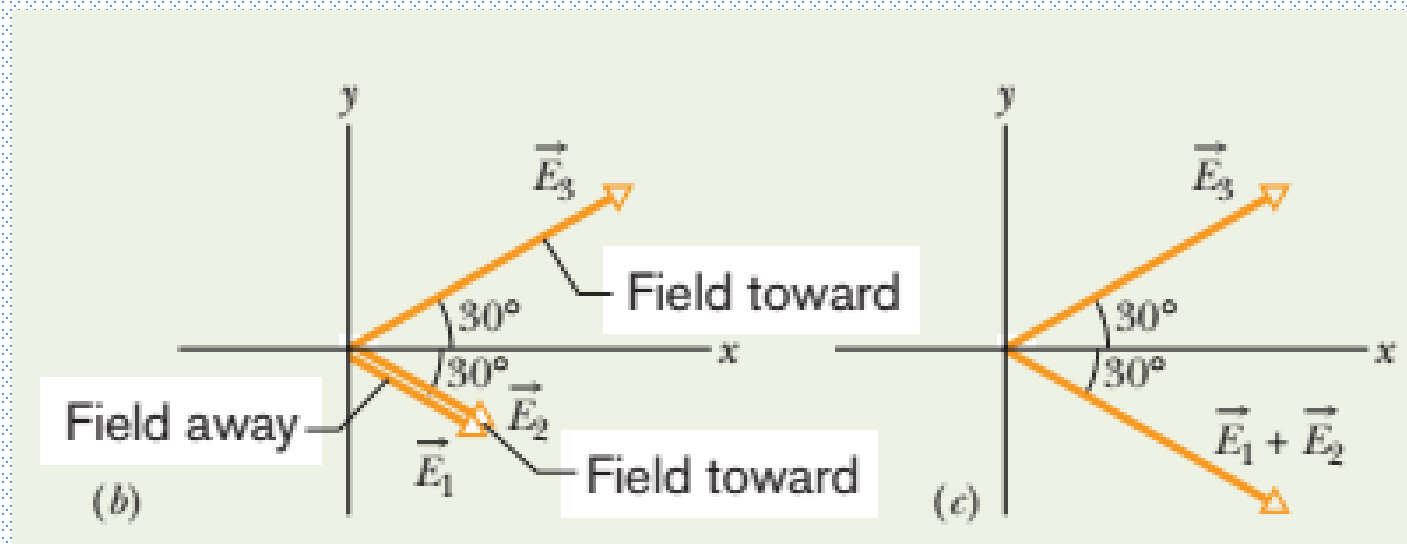
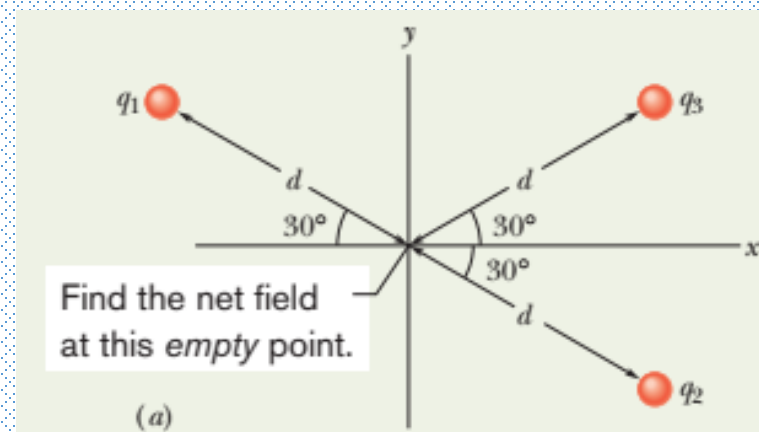


Fig. 22-6 The electric field vectors at various points around a positive point charge.

Sample Problem

Net electric field due to three charged particles

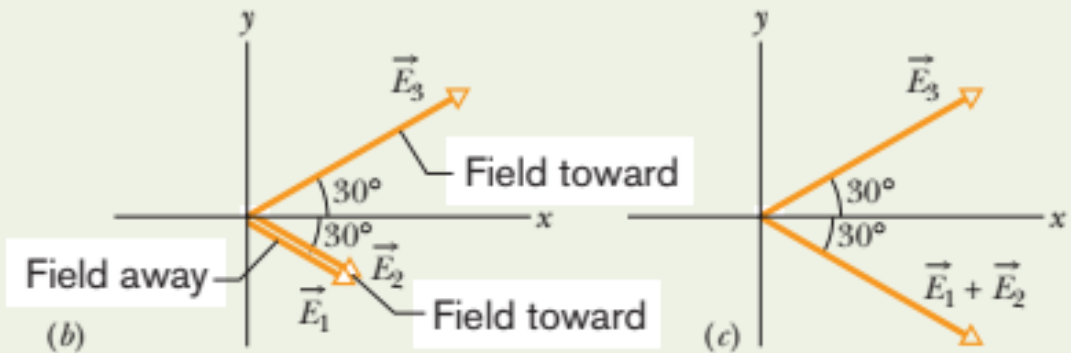
Figure 22-7a shows three particles with charges $q_1 = +2Q$, $q_2 = -2Q$, and $q_3 = -4Q$, each a distance d from the origin. What net electric field $\vec{E}$ is produced at the origin?



$$E_1 = \frac{1}{4\pi\epsilon_0} \frac{2Q}{d^2}.$$

$$E_2 = \frac{1}{4\pi\epsilon_0} \frac{2Q}{d^2} \quad \text{and} \quad E_3 = \frac{1}{4\pi\epsilon_0} \frac{4Q}{d^2}.$$

$$\begin{aligned} E_1 + E_2 &= \frac{1}{4\pi\epsilon_0} \frac{2Q}{d^2} + \frac{1}{4\pi\epsilon_0} \frac{2Q}{d^2} \\ &= \frac{1}{4\pi\epsilon_0} \frac{4Q}{d^2}, \end{aligned}$$



$$\begin{aligned} E &= 2E_{3x} = 2E_3 \cos 30^\circ \\ &= (2) \frac{1}{4\pi\epsilon_0} \frac{4Q}{d^2} (0.866) = \frac{6.93Q}{4\pi\epsilon_0 d^2}. \end{aligned} \quad (\text{Answer})$$

Table 22-2**Some Measures of Electric Charge**

Name	Symbol	SI Unit
Charge	q	C
Linear charge density	λ	C/m
Surface charge density	σ	C/m ²
Volume charge density	ρ	C/m ³

The Electric Field Due to an Electric Dipole

$$E = E_{(+)} - E_{(-)}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{q}{r_{(+)}^2} - \frac{1}{4\pi\epsilon_0} \frac{q}{r_{(-)}^2}$$

$$= \frac{q}{4\pi\epsilon_0(z - \frac{1}{2}d)^2} - \frac{q}{4\pi\epsilon_0(z + \frac{1}{2}d)^2}$$

$$E = \frac{q}{4\pi\epsilon_0 z^2} \left(\frac{1}{\left(1 - \frac{d}{2z}\right)^2} - \frac{1}{\left(1 + \frac{d}{2z}\right)^2} \right)$$

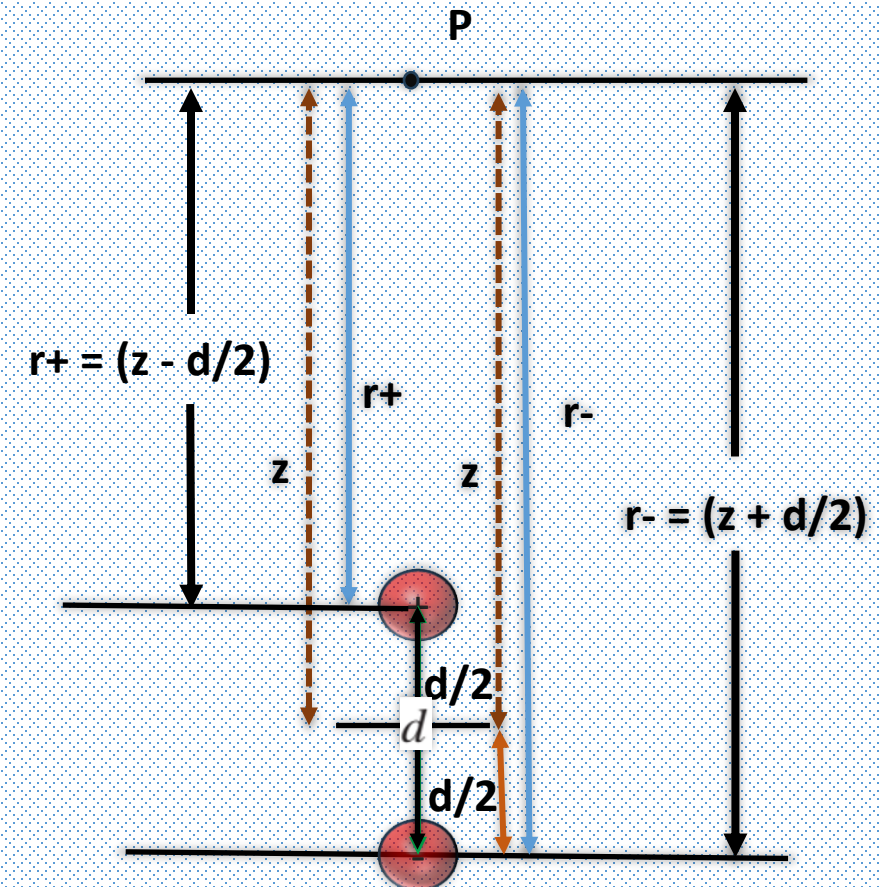
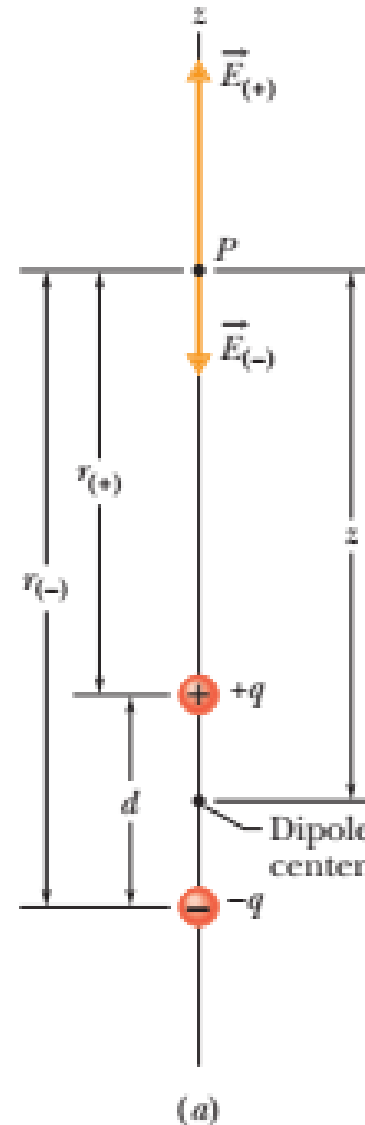
$$E = \frac{q}{4\pi\epsilon_0 z^2} \frac{2d/z}{\left(1 - \left(\frac{d}{2z}\right)^2\right)^2}$$

Up here the $+$ field dominates.



Down here the $-q$ field dominates.

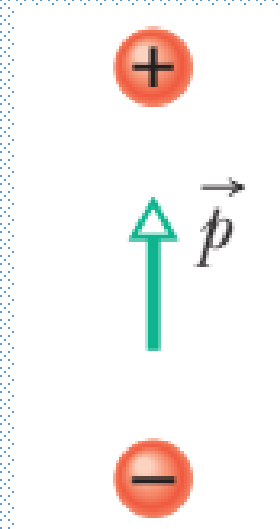
(b)



$$= \frac{q}{2\pi\epsilon_0 z^3} \frac{d}{\left(1 - \left(\frac{d}{2z}\right)^2\right)^2}.$$

$$d/2z \ll 1$$

$$E = \frac{1}{2\pi\epsilon_0} \frac{qd}{z^3}.$$

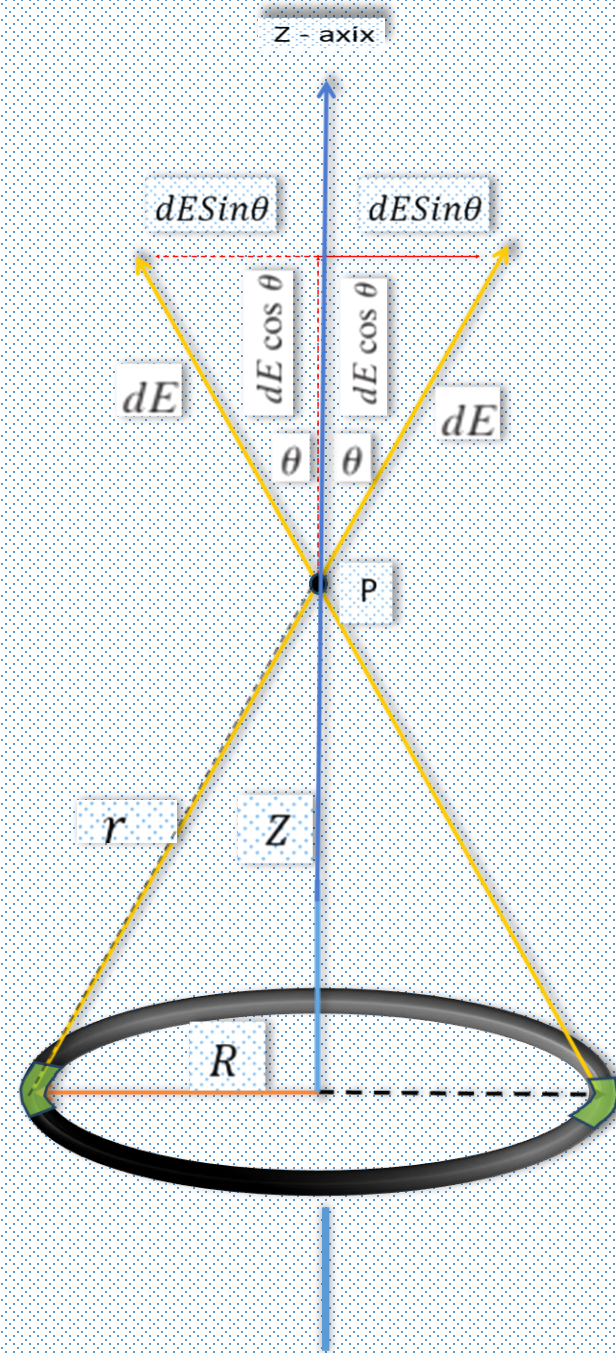
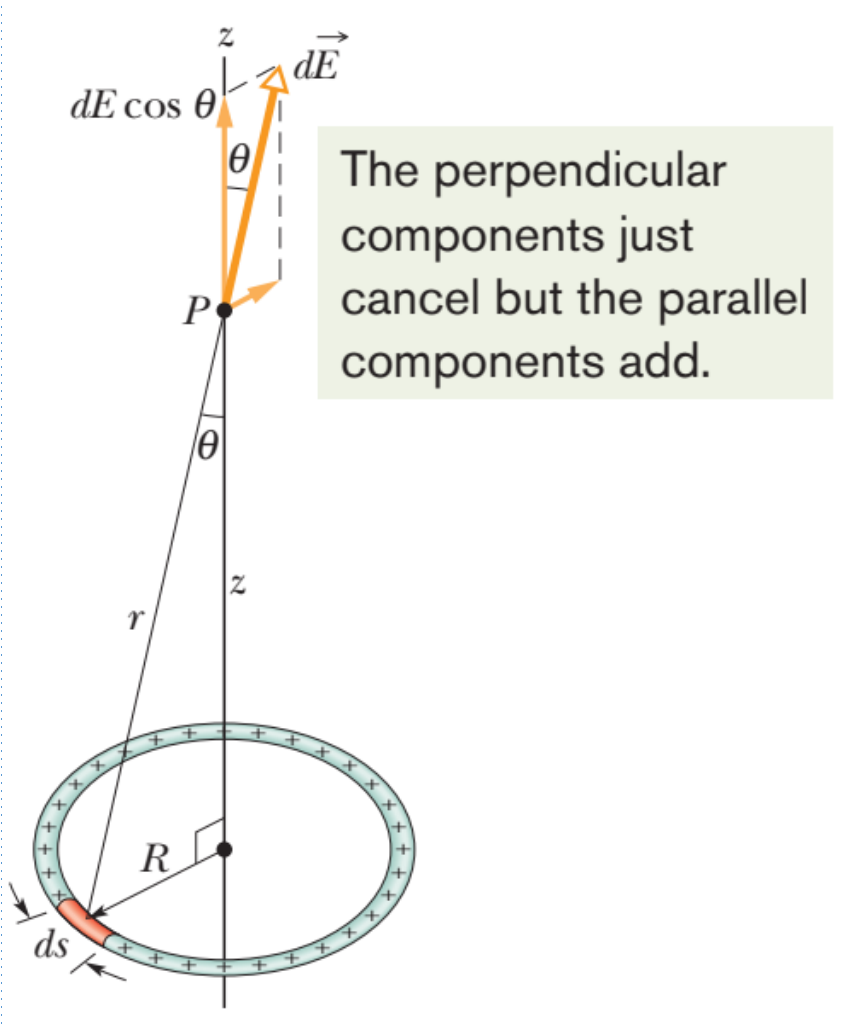


The product qd , which involves the two intrinsic properties q and d of the dipole, is the magnitude p of a vector quantity known as the **electric dipole moment** $\vec{p}$ of the dipole. (The unit of $\vec{p}$ is the coulomb-meter.) Thus, we can write Eq. 22-8 as

$$E = \frac{1}{2\pi\epsilon_0} \frac{p}{z^3} \quad (\text{electric dipole}). \quad (22-9)$$

The direction of $\vec{p}$ is taken to be from the negative to the positive end

Electric Field at a point due to ring of charges



22-6 The Electric Field Due to a Line of Charge

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2}$$

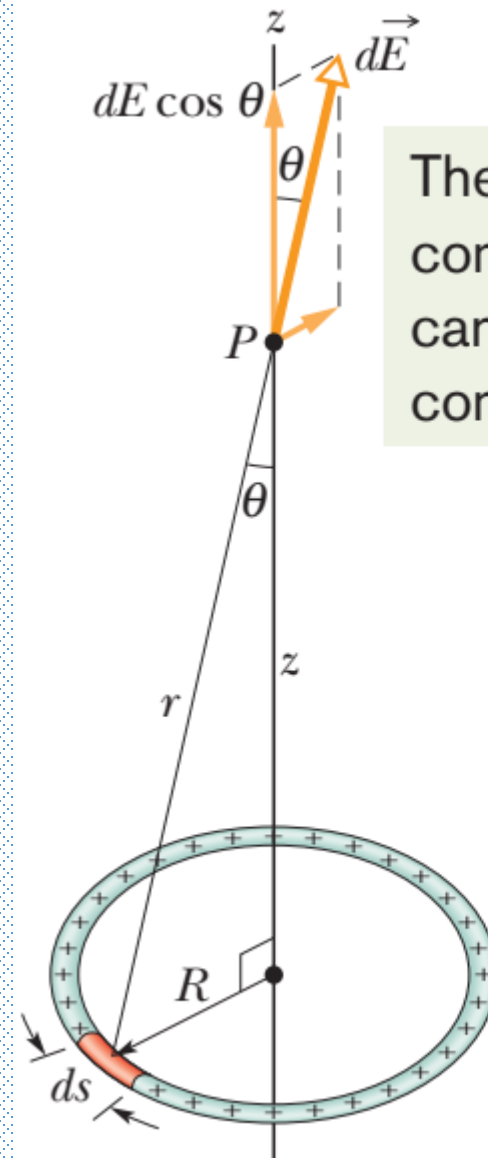
$$dq = \lambda ds.$$

$$= \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{r^2}.$$

$$dE = \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{(z^2 + R^2)}.$$

$$\cos \theta = \frac{z}{r} = \frac{z}{(z^2 + R^2)^{1/2}}.$$

$$dE \cos \theta = \frac{z\lambda}{4\pi\epsilon_0(z^2 + R^2)^{3/2}} ds.$$



The perpendicular components just cancel but the parallel components add.

$$E = \int dE \cos \theta = \frac{z\lambda}{4\pi\epsilon_0(z^2 + R^2)^{3/2}} \int_0^{2\pi R} ds$$

$$= \frac{z\lambda(2\pi R)}{4\pi\epsilon_0(z^2 + R^2)^{3/2}}.$$

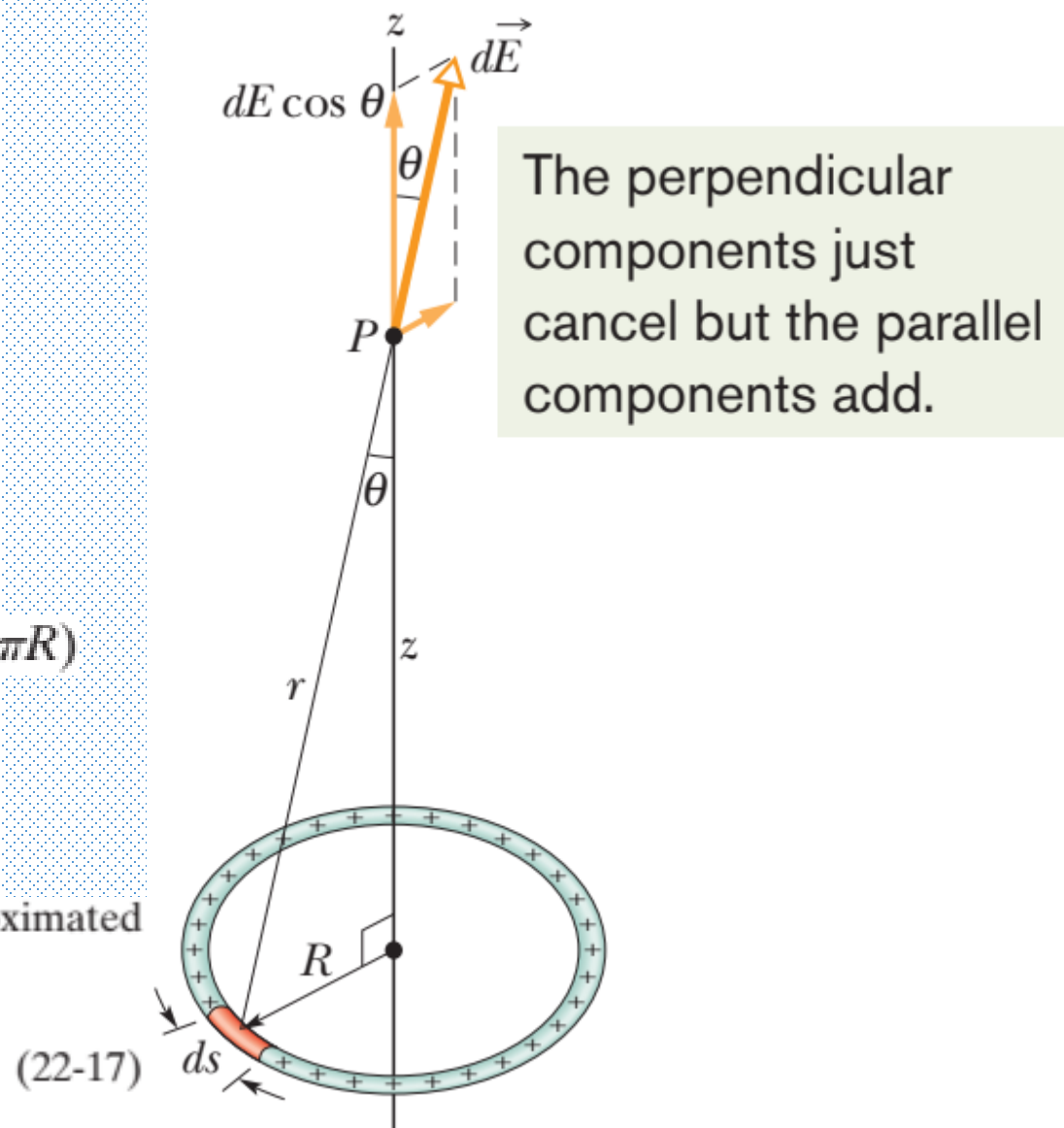
Since λ is the charge per length of the ring, the term $\lambda(2\pi R)$

$$E = \frac{qz}{4\pi\epsilon_0(z^2 + R^2)^{3/2}} \quad (\text{charged ring}).$$

$z \gg R$. For such a point, the expression $z^2 + R^2$ in Eq. 22-16 can be approximated as z^2 , and Eq. 22-16 becomes

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{z^2} \quad (\text{charged ring at large distance}).$$

$z = 0$. At that point, Eq. 22-16 tells us that $E = 0$.

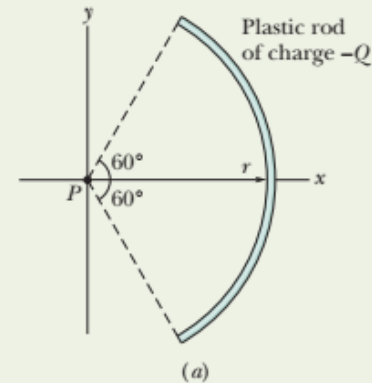


Sample Problem

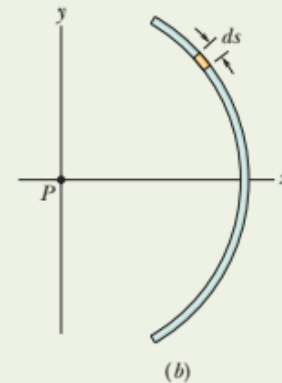
Electric field of a charged circular rod

Figure 22-11*a* shows a plastic rod having a uniformly distributed charge $-Q$. The rod has been bent in a 120° circular arc of radius r . We place coordinate axes such that the axis of symmetry of the rod lies along the x axis and the origin is at the center of curvature P of the rod. In terms of Q and r , what is the electric field $\vec{E}$ due to the rod at point P ?

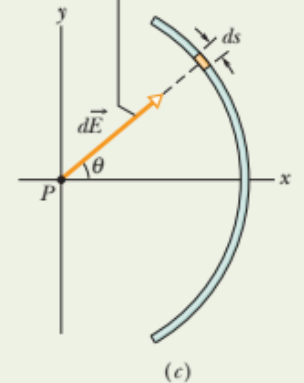
This negatively charged rod is obviously not a particle.



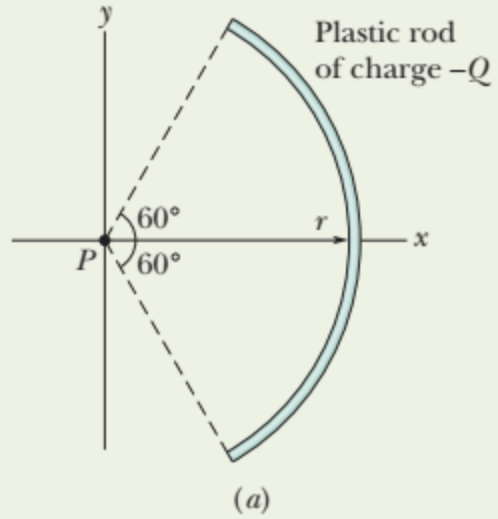
But we can treat this element as a particle.



Here is the field the element creates.



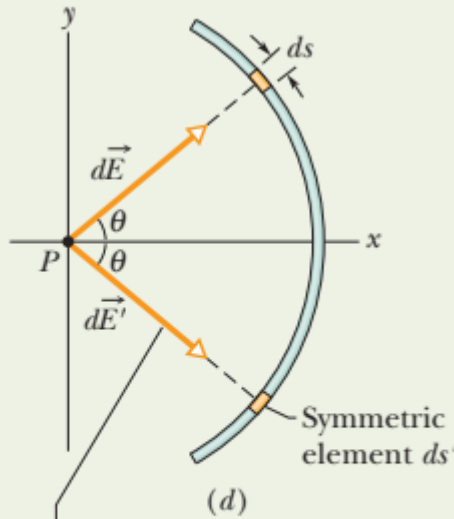
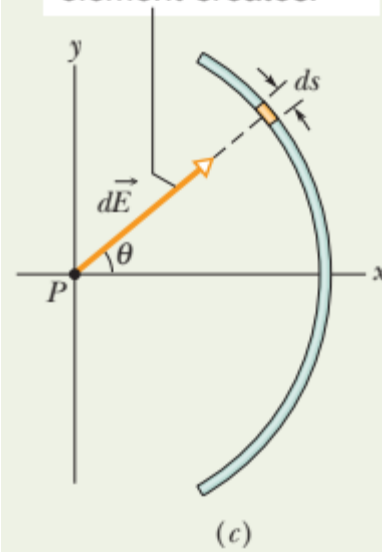
This negatively charged rod is obviously not a particle.



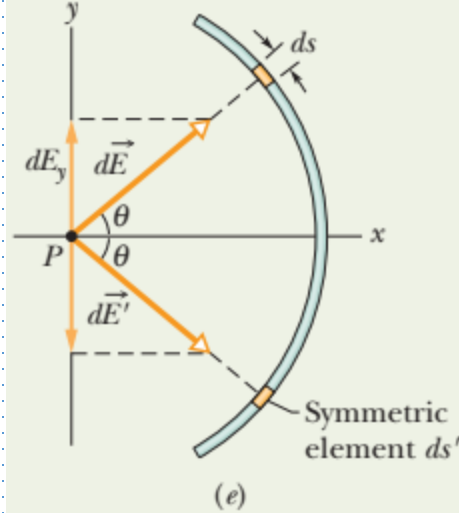
But we can treat this element as a particle.



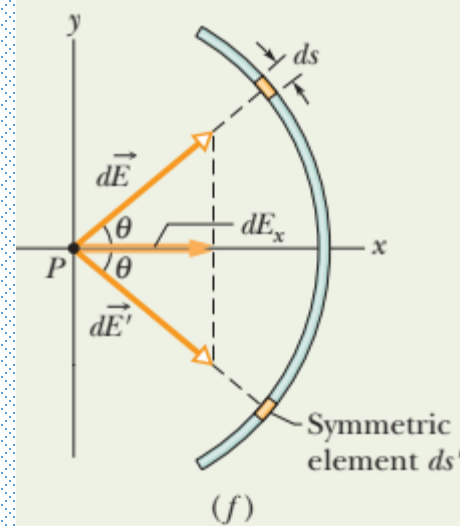
Here is the field the element creates.



Here is the field created by the symmetric element, same size and angle.



These y components just cancel, so neglect them.



These x components add. Our job is to add all such components.

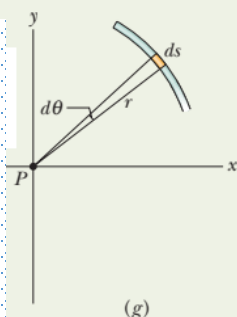
$$dq = \lambda ds. \quad (22-18)$$

The element's field: Our element produces a differential electric field $d\vec{E}$ at point P , which is a distance r from the element. Treating the element as a point charge, we can rewrite Eq. 22-3 to express the magnitude of $d\vec{E}$ as

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{r^2}. \quad (22-19)$$

$$dE_x = dE \cos \theta = \frac{1}{4\pi\epsilon_0} \frac{\lambda}{r^2} \cos \theta ds.$$

$$ds = r d\theta,$$



$$E = \int dE_x = \int_{-60^\circ}^{60^\circ} \frac{1}{4\pi\epsilon_0} \frac{\lambda}{r^2} \cos \theta r d\theta$$

$$= \frac{\lambda}{4\pi\epsilon_0 r} \int_{-60^\circ}^{60^\circ} \cos \theta d\theta = \frac{\lambda}{4\pi\epsilon_0 r} \left[\sin \theta \right]_{-60^\circ}^{60^\circ}$$

$$\begin{aligned} &= \frac{\lambda}{4\pi\epsilon_0 r} [\sin 60^\circ - \sin(-60^\circ)] \\ &= \frac{1.73\lambda}{4\pi\epsilon_0 r}. \end{aligned} \quad (22-21)$$

Charge density: To evaluate λ , we note that the rod subtends an angle of 120° and so is one-third of a full circle. Its arc length is then $2\pi r/3$, and its linear charge density must be

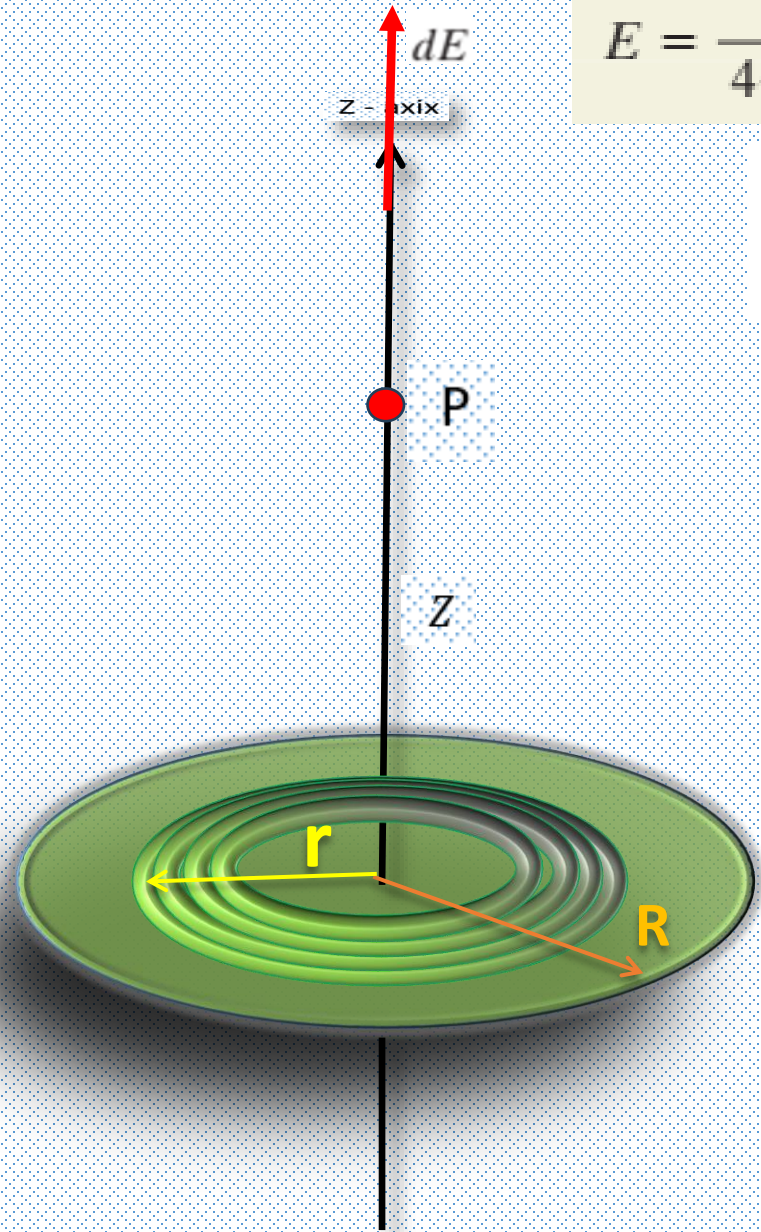
$$\lambda = \frac{\text{charge}}{\text{length}} = \frac{Q}{2\pi r/3} = \frac{0.477Q}{r}.$$

Substituting this into Eq. 22-21 and simplifying give us

$$\begin{aligned} E &= \frac{(1.73)(0.477Q)}{4\pi\epsilon_0 r^2} \\ &= \frac{0.83Q}{4\pi\epsilon_0 r^2}. \end{aligned} \quad (\text{Answer})$$

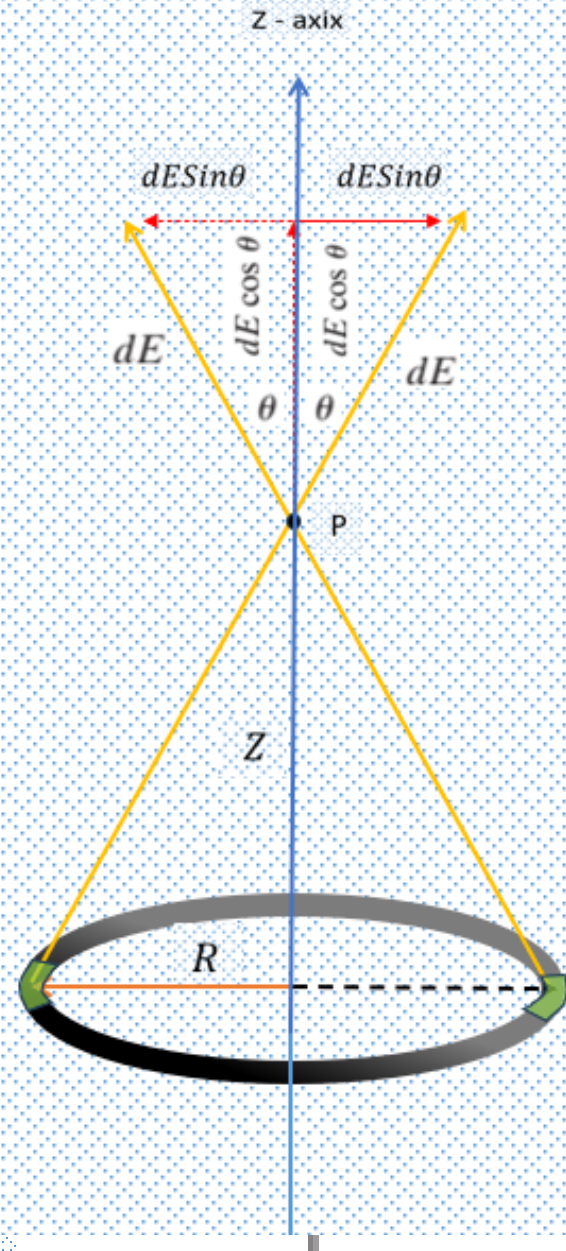
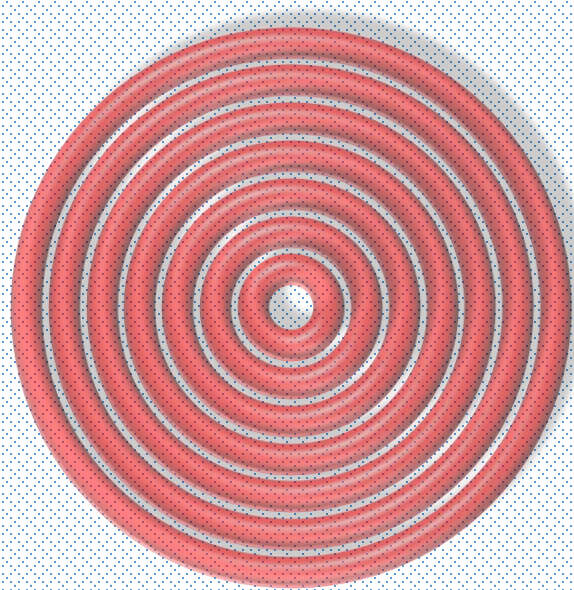
The direction of $\vec{E}$ is toward the rod, along the axis of symmetry of the charge distribution. We can write $\vec{E}$ in unit-vector notation as

$$\vec{E} = \frac{0.83Q}{4\pi\epsilon_0 r^2} \hat{i}.$$



$$E = \frac{qz}{4\pi\epsilon_0(z^2 + R^2)^{3/2}} \quad (\text{charged ring})$$

$$dE = \frac{z\sigma 2\pi r dr}{4\pi\epsilon_0(z^2 + r^2)^{3/2}}$$



22-7 The Electric Field Due to a Charged Disk

$$dq = \sigma dA = \sigma (2\pi r dr),$$

$$dE = \frac{z\sigma 2\pi r dr}{4\pi\epsilon_0(z^2 + r^2)^{3/2}}$$

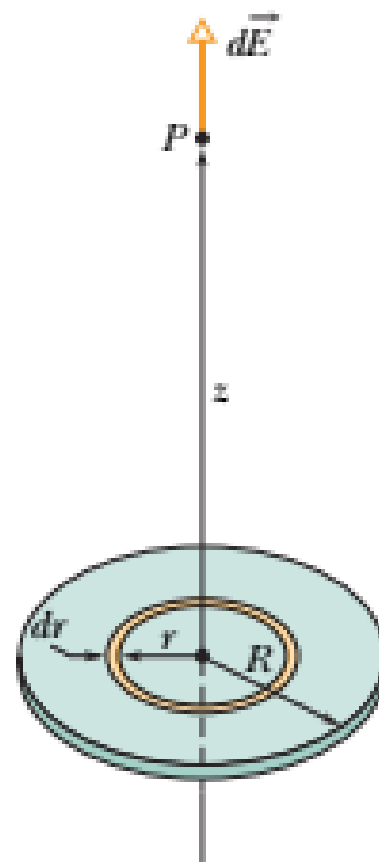
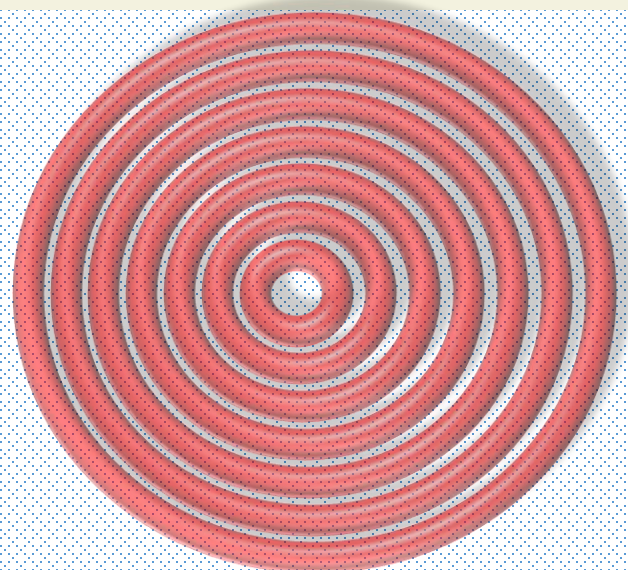
$$dE = \frac{\sigma z}{4\epsilon_0} \frac{2r dr}{(z^2 + r^2)^{3/2}}.$$

$$E = \int dE = \frac{\sigma z}{4\epsilon_0} \int_0^R (z^2 + r^2)^{-3/2} (2r) dr.$$

$$\int X^m dX = \frac{X^{m+1}}{m+1},$$

$$E = \frac{\sigma z}{4\epsilon_0} \left[\frac{(z^2 + r^2)^{-1/2}}{-\frac{1}{2}} \right]_0^R.$$

$$E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right) \quad (\text{charged disk})$$



If we let $R \rightarrow \infty$ while keeping z finite, the second term in the parentheses in Eq. 22-26 approaches zero, and this equation reduces to

$$E = \frac{\sigma}{2\epsilon_0} \quad (\text{infinite sheet}). \quad (22-27)$$

Sample Problem

Motion of a charged particle in an electric field

Figure 22-17 shows the deflecting plates of an ink-jet printer, with superimposed coordinate axes. An ink drop with a mass m of 1.3×10^{-10} kg and a negative charge of magnitude $Q = 1.5 \times 10^{-13}$ C enters the region between the plates, initially moving along the x axis with speed $v_x = 18$ m/s. The length L of each plate is 1.6 cm. The plates are charged and thus produce an electric field at all points between them. Assume that field $\vec{E}$ is downward directed, is uniform, and has a magnitude of 1.4×10^6 N/C. What is the vertical deflection of the drop at the far edge of the plates? (The gravitational force on the drop is small relative to the electrostatic force acting on the drop and can be neglected.)

$$y = \frac{1}{2}a_y t^2$$

$$L = v_x t,$$

$$y = \frac{QEL^2}{2mv_x^2}$$

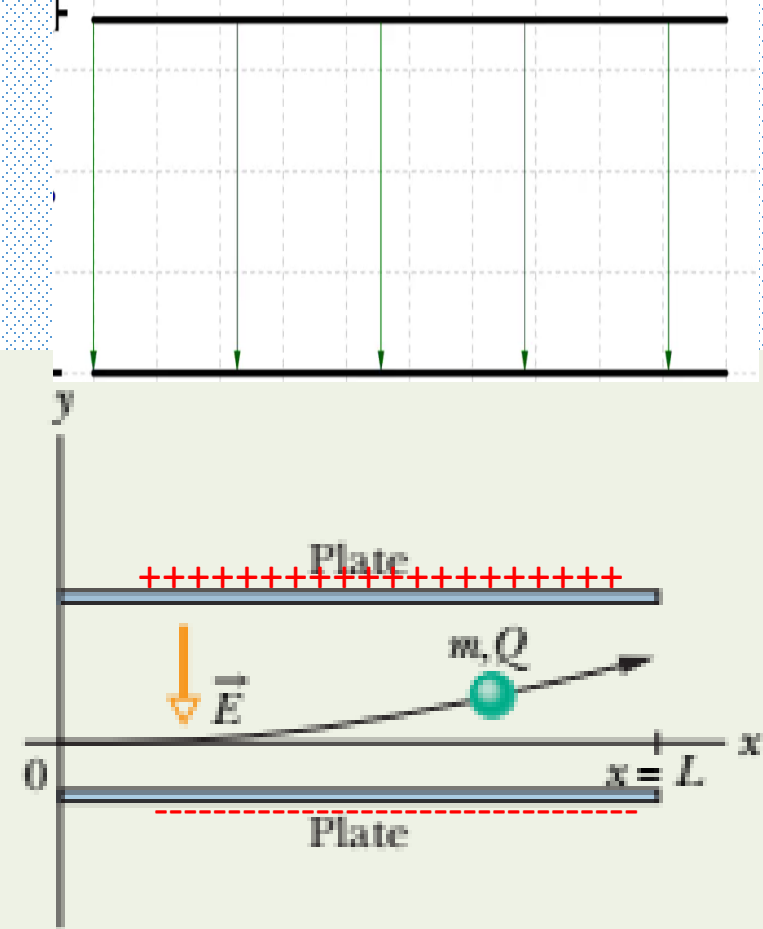
$$a_y = \frac{F}{m} = \frac{QE}{m}.$$

$$= \frac{(1.5 \times 10^{-13} \text{ C})(1.4 \times 10^6 \text{ N/C})(1.6 \times 10^{-2} \text{ m})^2}{(2)(1.3 \times 10^{-10} \text{ kg})(18 \text{ m/s})^2}$$

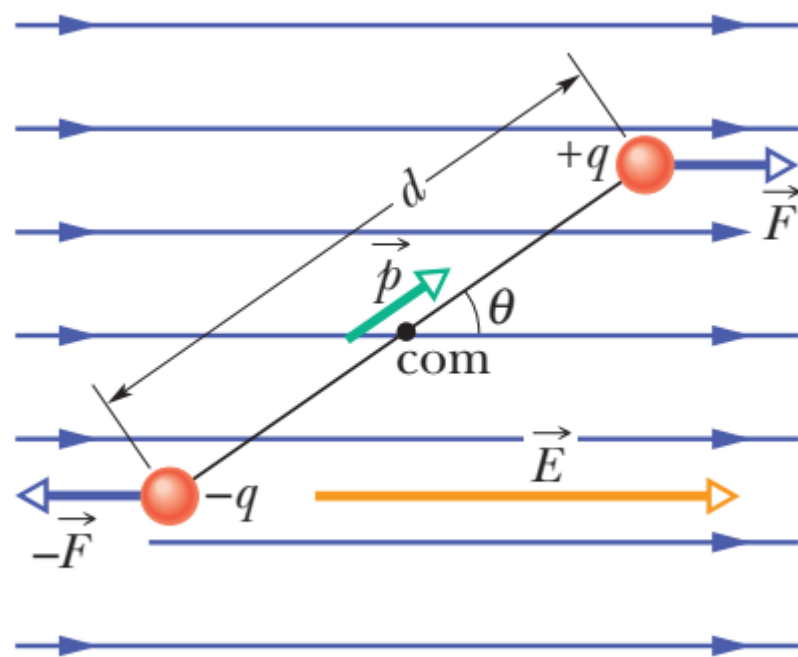
$$= 6.4 \times 10^{-4} \text{ m}$$

$$= 0.64 \text{ mm.}$$

(Answer)



22-9 A Dipole in an Electric Field



(a)

The dipole is being torqued into alignment.



(b)

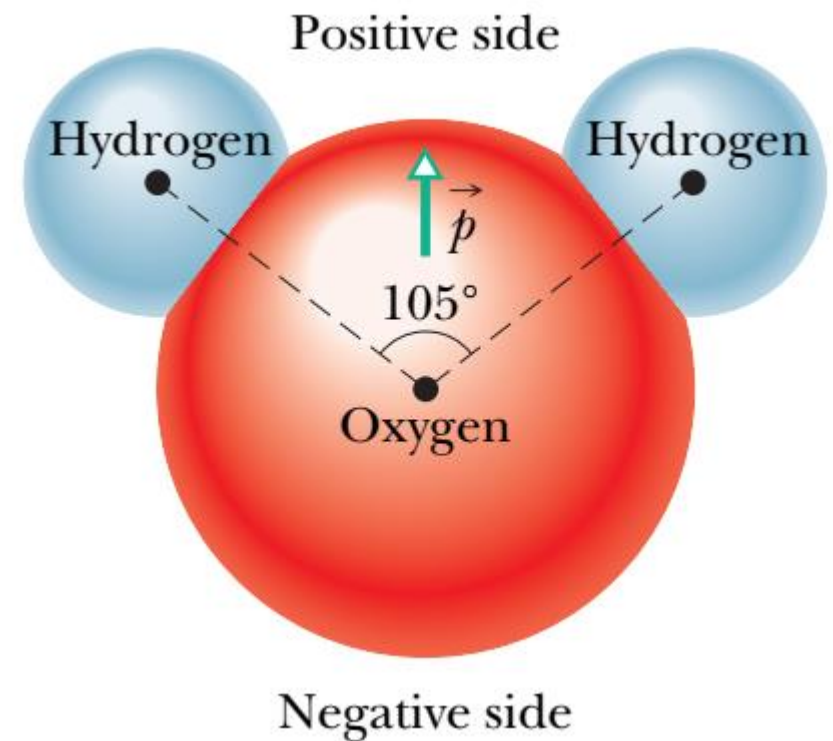


Fig. 22-18 A molecule of H_2O , showing the three nuclei (represented by dots) and the regions in which the electrons can be located. The electric dipole moment $\vec{p}$ points from the (negative) oxygen side to the (positive) hydrogen side of the molecule.

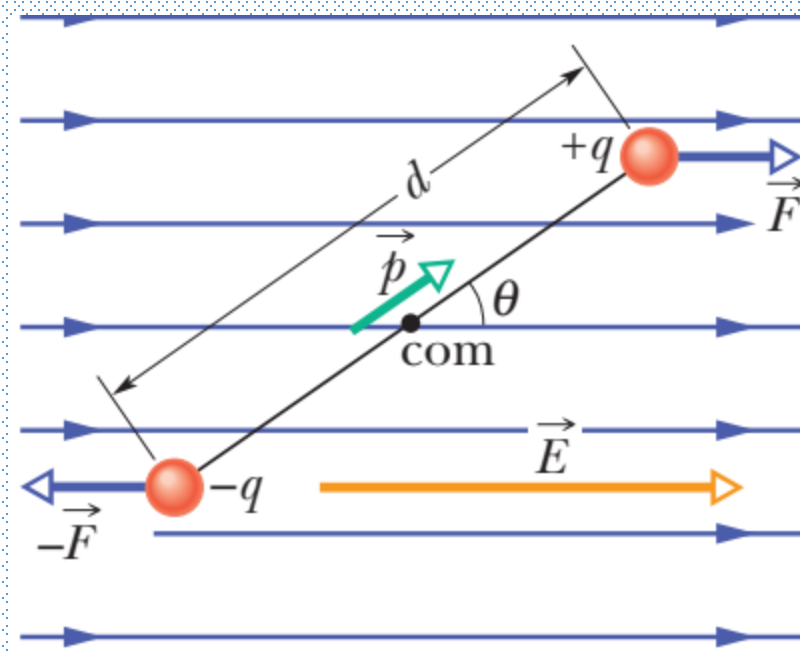
the magnitude of the net torque $\vec{\tau}$ as

$$\tau = Fx \sin \theta + F(d - x) \sin \theta$$

$$= Fd \sin \theta. \text{ substitute } qE \text{ for } F$$

$$p = qd.$$

$$\tau = pE \sin \theta.$$

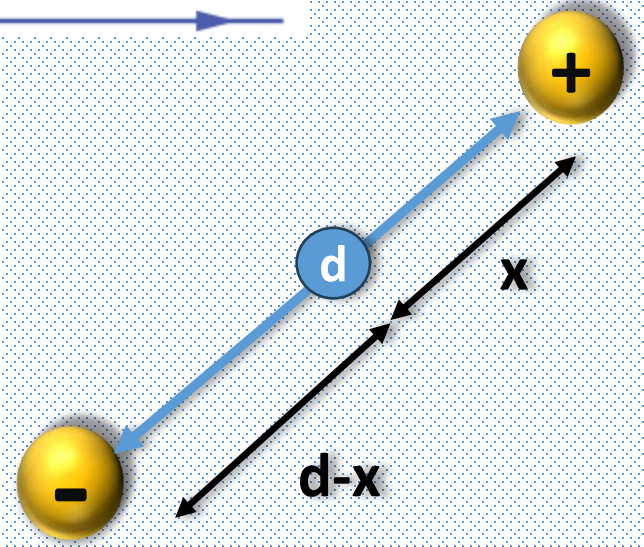


We can generalize this equation to vector form as

$$\vec{\tau} = \vec{p} \times \vec{E} \quad (\text{torque on a dipole}).$$

$$\tau = -pE \sin \theta.$$

clockwise rotation by including a minus sign



Potential Energy of an Electric Dipole

Potential energy can be associated with the orientation of an electric dipole

The dipole has its least potential energy when it is in its equilibrium which is when its moment $\vec{p}$ is lined up with the field $\vec{E}$

$$\vec{\tau} = \vec{p} \times \vec{E} = 0$$

$$U = -\vec{p} \cdot \vec{E} \quad (\text{potential energy of a dipole}).$$

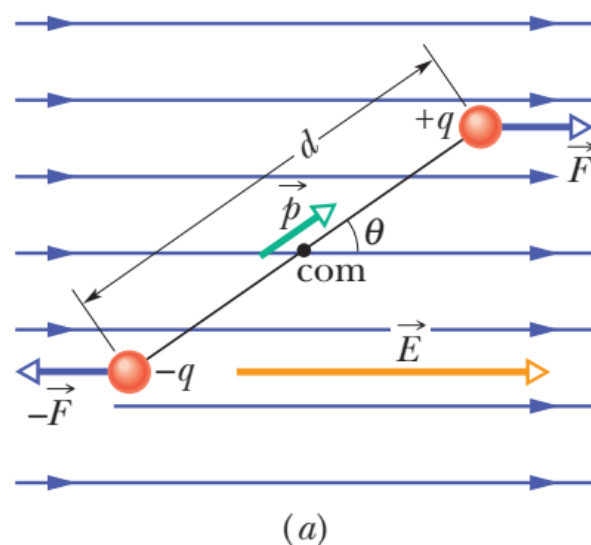
$$U = -W \quad W = \int \tau d\theta$$

$$W = -\Delta U = -(U_f - U_i),$$

$$U = -W = -\int_{90^\circ}^{\theta} \tau d\theta \\ = \int_{90^\circ}^{\theta} pE \sin \theta d\theta.$$

Evaluating the integral leads to

$$U = -pE \cos \theta.$$



••31 SSM ILW WWW In Fig. 22-49, a nonconducting rod of length $L = 8.15$ cm has a charge $-q = -4.23$ fC uniformly distributed along its length. (a) What is the linear charge density of the rod? What are the (b) magnitude and (c) direction (relative to the positive direction of the x axis) of the electric field produced at point P , at distance $a = 12.0$ cm from the rod? What is the electric field magnitude produced at distance $a = 50$ m by (d) the rod and (e) a particle of charge $-q = -4.23$ fC that replaces the rod?

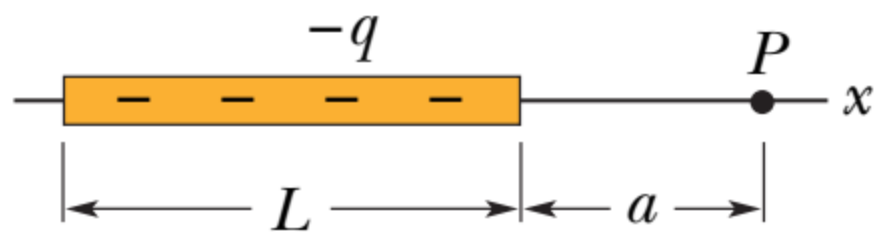
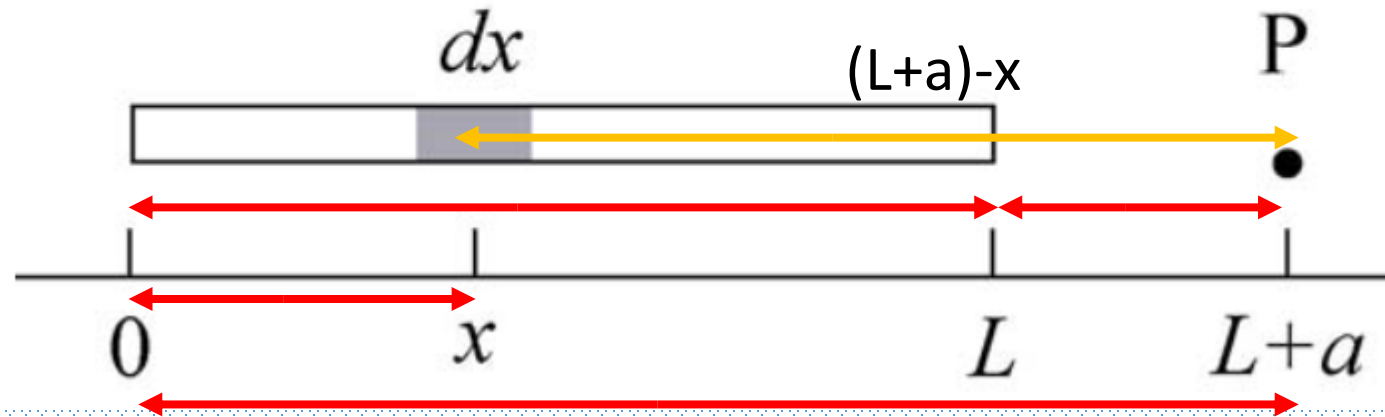


Fig. 22-49 Problem 31.

31. (a) The linear charge density is the charge per unit length of rod. Since the charge is uniformly distributed on the rod,

$$\lambda = \frac{-q}{L} = \frac{-4.23 \times 10^{-15} \text{ C}}{0.0815 \text{ m}} = -5.19 \times 10^{-14} \text{ C/m}.$$

(b) We position the x axis along the rod with the origin at the left end of the rod, as shown in the diagram.



Let dx be an infinitesimal length of rod at x . The charge in this segment is $dq = \lambda \, dx$. The charge dq may be considered to be a point charge. The electric field it produces at point P has only an x component, and this component is given by

$$dE_x = \frac{1}{4\pi\epsilon_0} \frac{\lambda \, dx}{(L + a - x)^2}.$$

$$\begin{aligned} E_x &= \frac{\lambda}{4\pi\epsilon_0} \int_0^L \frac{dx}{(L + a - x)^2} = \frac{\lambda}{4\pi\epsilon_0} \frac{1}{L + a - x} \Big|_0^L = \frac{\lambda}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{L + a} \right) \\ &= \frac{\lambda}{4\pi\epsilon_0} \frac{L}{a(L + a)} = -\frac{1}{4\pi\epsilon_0} \frac{q}{a(L + a)}, \end{aligned}$$

upon substituting $-q = \lambda L$. With $q = 4.23 \times 10^{-15}$ C, $L = 0.0815$ m and $a = 0.120$ m, we obtain $E_x = -1.57 \times 10^{-3}$ N/C, or $|E_x| = 1.57 \times 10^{-3}$ N/C.

(c) The negative sign in E_x indicates that the field points in the $-x$ direction, or -180° counterclockwise from the $+x$ axis.

(d) If a is much larger than L , the quantity $L + a$ in the denominator can be approximated by a , and the expression for the electric field becomes

$$E_x = -\frac{q}{4\pi\epsilon_0 a^2}.$$

Since $a = 50 \text{ m} \gg L = 0.0815 \text{ m}$, the above approximation applies, and we have $E_x = -1.52 \times 10^{-8} \text{ N/C}$, or $|E_x| = 1.52 \times 10^{-8} \text{ N/C}$.

(e) For a particle of charge $-q = -4.23 \times 10^{-15} \text{ C}$, the electric field at a distance $a = 50 \text{ m}$ away has a magnitude $|E_x| = 1.52 \times 10^{-8} \text{ N/C}$.